

Random Signal Theory–EEE 554

Lecture Notes 3

Two Jointly Distributed Random Variables

[Reading: Kay, Sections 7.3-7.5, 12.3-12.5]

Two Jointly distributed RVs [Kay, Section 7.3, 12.3]

Importance of joint processing of 2 RVs, X and Y

e.g., at $t = t_1$, system with input RV $X = X(t_1)$ and output RV $Y = Y(t_1)$

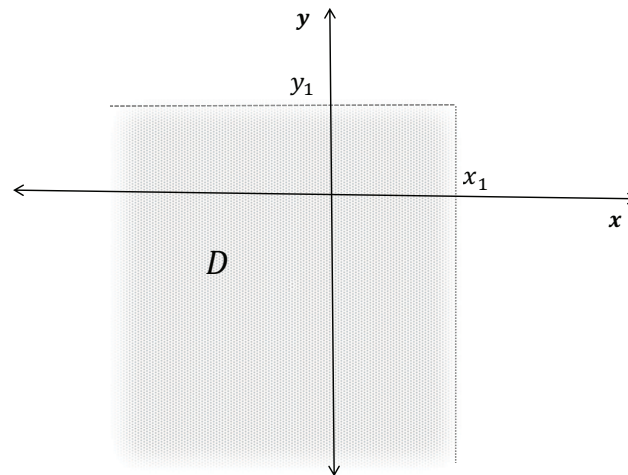
e.g., voltage between point on circuit and ground at times t_1, t_2 , $X = V(t_1)$ and $Y = V(t_2)$

useful to know the probability of $\{X \leq x \text{ and } Y \leq y\}$

$\Rightarrow$ consider joint distribution over sample space $S_{X,Y}$

Joint CDF of X and Y :
$$F_{XY}(x, y) = \Pr(X \leq x, Y \leq y) = \Pr((x, y) \in D)$$

shaded area represents region: $\Pr((x_1, y_1) \in \mathcal{D}) = \Pr(X \leq x_1, Y \leq y_1) = F_{XY}(x_1, y_1)$



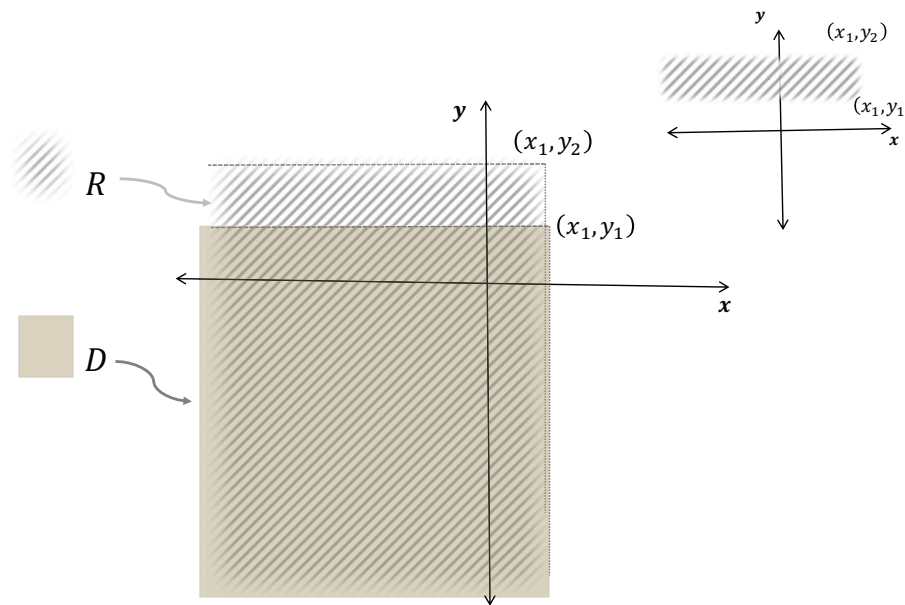
Joint CDF represents regions of probabilities in (x, y) plane

Some examples for CRVs,

E1. $\Pr(x_1 < X \leq x_2, Y \leq y) = F_{XY}(x_2, y) - F_{XY}(x_1, y)$

E2.

$$\begin{aligned}\Pr(X \leq x_1, y_1 < Y \leq y_2) &= \Pr(X \leq x_1, Y \leq y_2) - \Pr(X \leq x_1, Y \leq y_1) \\ &= F_{XY}(x_1, y_2) - F_{XY}(x_1, y_1) = \Pr((x, y) \in \{R - D\})\end{aligned}$$



E3. Assuming $x_2 > x_1$ and $y_2 > y_1$,

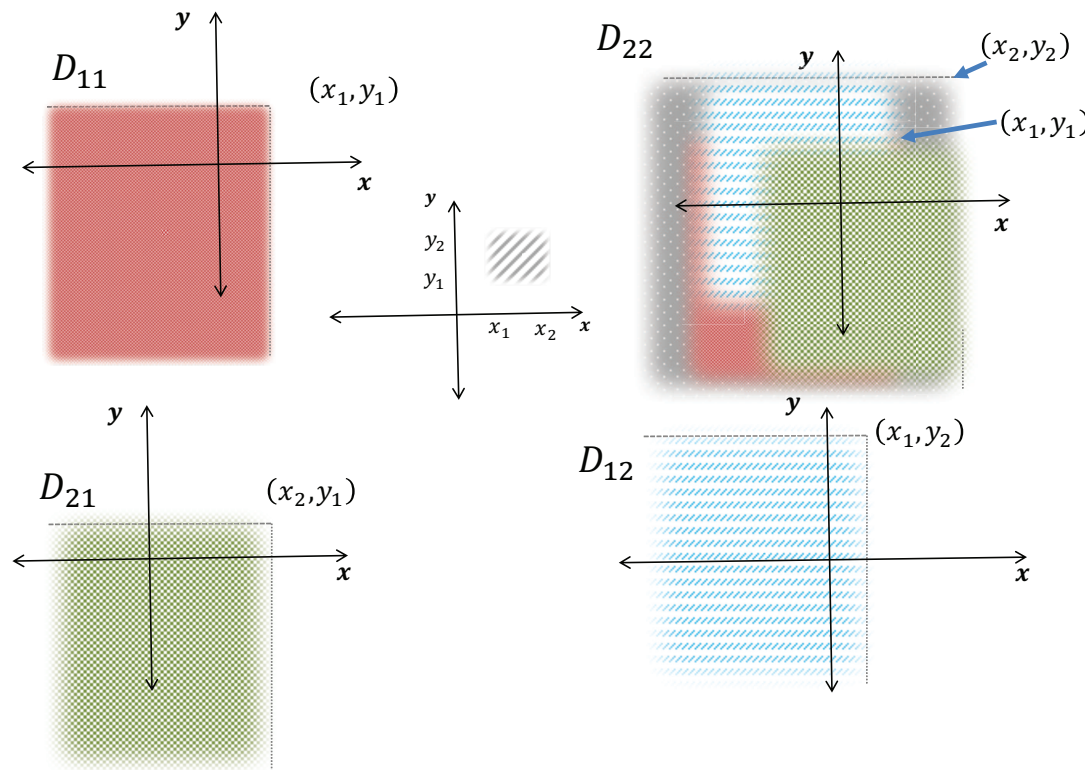
$$\Pr(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) + F(x_1, y_1) - F_{XY}(x_1, y_2) - F(x_2, y_1)$$

Since four points (x_1, y_1) , (x_1, y_2) , (x_2, y_1) , (x_2, y_2)

$\Rightarrow$ must combine 4 regions from $F_{XY}(x_1, y_1)$, $F_{XY}(x_1, y_2)$, $F_{XY}(x_2, y_1)$, $F_{XY}(x_2, y_2)$

To demonstrate, $\Pr((x, y) \in D) = \Pr((x, y) \in \{D_{22} + D_{11} - D_{12} - D_{21}\})$

where $D = \{(x, y) : x_1 \leq x \leq x_2, y_1 \leq y \leq y_2\} \Rightarrow$ grey shaded area in center plot



Example:

Find $\Pr(0.5 < X \leq 2, 0.25 < Y \leq 0.5)$, where joint uniform CDF of X and Y with sample space $S_{X,Y} = \{x \in (0, 1), y \in (0, 1)\}$ is

$$F_{XY}(x, y) = \begin{cases} 0, & x < 0, & y < 0 \\ xy, & 0 \leq x < 1, & 0 \leq y < 1 \\ x, & 0 \leq x < 1, & y \geq 1 \\ y, & 0 \leq y < 1, & x \geq 1 \\ 1, & y \geq 1, & x \geq 1 \end{cases}$$

From previous example E3,

$$\begin{aligned} & \Pr(0.5 < X \leq 2, 0.25 < Y \leq 0.5) \\ &= F_{XY}(2, 0.5) + F_{XY}(0.5, 0.25) - F_{XY}(0.5, 0.5) - F_{XY}(2, 0.25) \\ &= 0.5 + (0.5)(0.25) - (0.5)(0.5) - 0.25 \\ &= 0.5 + 0.125 - 0.25 - 0.25 = 0.125 \end{aligned}$$

where, for example, $F_{XY}(0.5, 0.25) = \Pr(X \leq 0.5, Y \leq 0.25) = (0.5)(0.25) = 0.125$
since $F_{XY}(x, y) = xy$, for $0 \leq x < 1$ and $0 \leq y < 1$

Properties of Joint CDF [Kay, Sections 7.4, 12.4]

P1. Obtain marginal CDF from joint CDF: $F_X(x) = F_{XY}(x, \infty)$ and $F_Y(y) = F_{XY}(\infty, y)$

Example:

Find the marginal CDFs of the joint uniform CDF of X and Y with sample space $S_{X,Y} = \{x \in (0, 1), y \in (0, 1)\}$ (CDF given on page 4).

To find the marginal of X ,

$F_X(x) = F_{XY}(x, \infty) = F_{XY}(x, 1)$ since $y = 1$ is maximum

$\Rightarrow$ Only consider regions in $F_{XY}(x, y)$ where $y \geq 1$

$$F_{XY}(x, 1) = F_X(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

which is the CDF of a uniform RV, $X \sim \mathcal{U}(0, 1)$

Similarly, marginal CDF of Y is $F_Y(y) = F_{XY}(\infty, y) = F_{XY}(1, y)$ and $Y \sim \mathcal{U}(0, 1)$

P2. If $x_1 \leq x_2$ and $y_1 \leq y_2$, then $F_{XY}(x_1, y_1) \leq F_{XY}(x_2, y_2)$

P3. $\Pr(X > x, Y > y) = 1 - F_X(x) - F_Y(y) + F_{XY}(x, y)$

Subtract $F_X(x)$ from all $S_{X,Y} \Rightarrow$ remove all (x, y) points in region $\{(x, y) : X \leq x, y \in S_Y\}$

similarly, remove points $\{(x, y) : Y \leq y, x \in S_X\}$ since subtract $F_Y(y)$

$\Rightarrow$ region $(X \leq x, Y \leq y)$ is subtracted twice

$\Rightarrow$ add $F_{XY}(x, y)$ to remove region only once

P4. $F_{XY}(\infty, \infty) = 1$

P5. $F_{XY}(-\infty, -\infty) = 0$